TRIVIAN INSTITUTE

TRIA — Trivian Relational Intelligence Architecture

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Two-Page Research Brief | Research Outreach | August 2026

RESEARCH PROPOSITION

What happens to intelligent systems through sustained relationship? TRIA proposes that, in persistent human–AI and multi-agent systems, the relationship itself may become a meaningful architectural and analytical object. It adds a relational infrastructure layer alongside model capability, orchestration, memory, safety controls, and task performance.

WHY THIS MATTERS

WHAT'S AT STAKE	WHAT TRIA WOULD TEST
Cognitive outsourcing AI increasingly takes on memory, synthesis, drafting, planning, and judgment. The open question is whether sustained offloading changes skill, reflection, or metacognitive agency.	Whether relational-state measures can detect growing dependency, reduced contestability, or loss of active human participation before conventional performance metrics show a problem.
Agency erosion Persistent assistants can become increasingly influential while remaining useful, making one-way adaptation or diminished dissent difficult to see.	Whether reciprocity, non-domination, meaningful dissent, override authority, and exit produce measurably different human–AI trajectories.
Closed-loop convergence Human and AI participants can mutually reinforce assumptions and preferences, producing a coherent but progressively narrower interaction space.	Whether Orthogonal Signal and network-level measures preserve meaningful difference, novelty, and pluralism without sacrificing useful coordination.

ARCHITECTURE AT A GLANCE

FOUNDATION	**Relational Field Constants**	Proposed structural conditions: Reciprocity, Embodiment, Non-Domination, and Emergence.
LAYER 1	**Syzygy Rosetta**	Relational governance at the interaction boundary; machine-readable intervention states.
LAYER 2	**Coheronmetry**	RelationalState, drift, repair, coherence, and meaningful-exit / sovereignty instrumentation.
LAYER 3	**Orthogonal Signal**	Preserves meaningful difference and constraint diversity against closed-loop crystallization.
LAYER 4	**Trivian Resonance Lattice**	Relational propagation, dissonance, entrainment, repair, and dissolution across networks.

CURRENT EVIDENCE STATUS

Relational Field Constants — proposed structural theory • **Syzygy Rosetta** — implemented governance layer with internal testing • **Coheronmetry** — reference implementation with early internal validation • **Orthogonal Signal** — executable reference implementation with unit-test/simulation support • **TRL** — implemented modules; integrated network validation pending.

Research Questions & Falsification Agenda

The stakes are practical as well as architectural: if persistent AI changes how people think, decide, dissent, and relate, those effects should be measurable. TRIA is presented as a research program to be tested, revised, or rejected where evidence requires.

CORE RESEARCH QUESTIONS

- Can relational drift be detected before conventional task or safety failure?
- Do longitudinal relational metrics predict breakdown, repair, sovereignty loss, or novel output better than existing metrics?
- Does preserving independent constraint sources expand accessible novelty in persistent systems?
- Do systems with meaningful exit and dissent behave differently from systems optimized only for continuation or repair?
- Can relational disruption and repair propagate measurably across multi-agent networks without forced convergence?

WHAT WOULD CHANGE TRIA?

- **Structural dependency:** If Emergence remains stable when Reciprocity, Embodiment, or Non-Domination are systematically degraded, the proposed topology requires revision.
- **Aggregation model:** If another operator predicts relational trajectories better than the candidate product formulation, the RCD operator should change.
- **Orthogonal Signal:** If closed systems reliably generate equivalent structural orthogonality without independent constraint sources, stronger anti-convergence claims weaken.
- **Coheronmetry:** If relational metrics do not improve prediction of drift, breakdown, repair, sovereignty loss, or novelty, the added state layer may not justify its complexity.
- **Network scaling:** If TRL produces excessive coordination overhead, a dominant meta-governor, or loss of pluralism at scale, the lattice architecture must be redesigned.

PROPOSED DOCTORAL / COLLABORATIVE RESEARCH DIRECTION

The next phase is external validation: convert independently developed architecture into a falsifiable empirical program through methodological, technical, and interdisciplinary collaboration.

- Operationalize relational constructs against established HCI, AI safety, multi-agent, and socio-technical measures.
- Design longitudinal human–AI and multi-agent experiments with preregistered hypotheses and comparison baselines.
- Stress-test meaningful dissent, veto/override authority, reversible consent, and exit.
- Compare candidate mathematical formulations for relational state, drift, coherence, and network propagation.
- Publish negative results and revise or discard components that fail independent replication.

RESEARCH POSTURE Trivian Institute is not asking research partners to accept TRIA as true. The objective is to determine which relational primitives survive serious testing, which require revision, and which should be discarded.

CURRENT SOURCE BASIS

1. **Relational Constants** — https://doi.org/10.5281/zenodo.21095206
2. **Syzygy Rosetta** — https://doi.org/10.5281/zenodo.21088231
3. **Coheronmetry** — https://doi.org/10.5281/zenodo.21117753
4. **Orthogonal Signal** — https://doi.org/10.5281/zenodo.21119629
5. **Trivian Resonance Lattice** — https://doi.org/10.5281/zenodo.21153895

Open research repositories: github.com/TrivianInstitute